London Ambulance Service NHS
NHS Trust

Medical and Surgical Conditions

Adrenal Insufficiency and Addison's Disease

Adrenal Insufficiency/Addison's Disease

Objective

Describe the nature and characteristics of Adrenal insufficiency

and

Explain how to manage a patient suffering Addisonians crisis

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Adrenal Insufficiency/Addison's Disease

Prevalence in the UK

Autoimmune Addisons affects 1 in 10,000

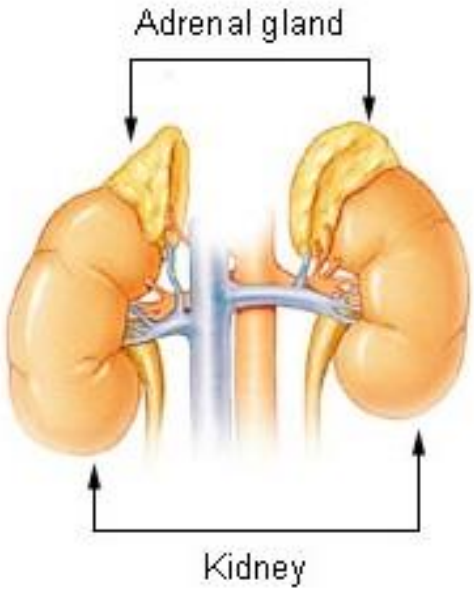
8,400 diagnosed cases

Usually young and middle aged people

(Ambulance Care Practice, 2016).

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Adrenal Insufficiency/Addison's Disease



Adrenal Insufficiency is a rare, potentially deadly but treatable disease.

In an acute crisis, the patient's health can deteriorate rapidly and result in death

Caroline 2014

Adrenal Insufficiency/Addison's Disease

Primary

Also called Addison's disease.

Damage to the adrenal cortex causes them to not be able to make enough cortisol and aldosterone.

Secondary

Caused by a de-functioning pituitary gland not signalling the adrenal glands to make cortisol.

Can also be caused by long term and / or large dosages of Steroid intake.

Caroline 2014

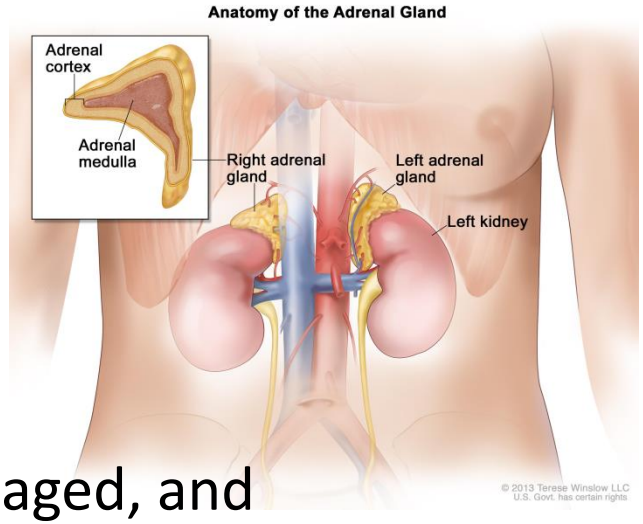
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Adrenal Insufficiency/Addison's Disease

Addison's disease is also known as primary adrenal insufficiency

The adrenal glands are two small glands that sit on top of the kidneys. They produce two essential hormones:

- Cortisol
- Aldosterone.



In Addison's disease, the adrenal gland is damaged, and not enough cortisol and aldosterone are produced.

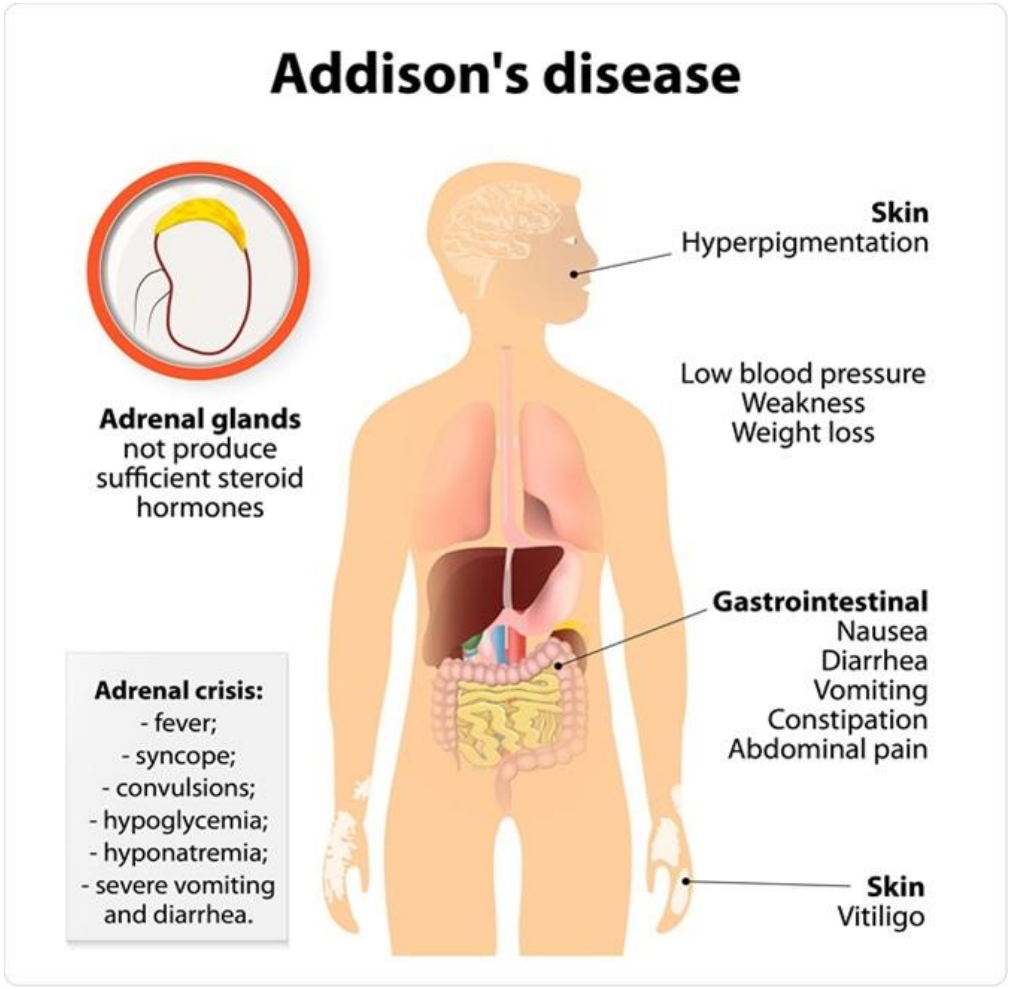
Caroline 2014

Adrenal Insufficiency/Addison's Disease

Cortisol and Aldosterone

- Help regulate blood pressure
- Help the immune system
- Balances the effect of insulin
- Assists the body responding to stress
- Regulates salts in the blood

Adrenal Insufficiency/Addison's Disease





Hyperpigmentation

Vitiligo



Adrenal Insufficiency/Addison's Disease

Hyperpigmentation





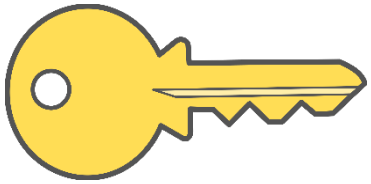


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Adrenal Insufficiency/Addison's Disease

Cause	Incidence %
Vomiting	33
Diarrhoea	23
Flu-like Illness	11
Major Infection	6
Surgery	6
Blackout/LOC	6
Injury/Severe Pain	4
Shock	4
Other	3
Severe fatigue/inadequate medication	2
Unknown	1
Anxiety/phsycological upset/stress	1

History is



Ambulance Care Practice, 2016

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Adrenal Insufficiency/Addison's Disease

Tablet or IM steroid treatment:

- 40 mg for more than a week.
- 20 mg or more for two weeks.
- 5 mg or more for more than 4 weeks.
- This can affects the body's own ability to produce Cortisol.
- This effect can last as long as a year after steroid discontinuation and increase the risk of developing adrenal crisis.



Fields,2009

Adrenal Insufficiency/Addison's Disease

Possible triggers of acute exacerbation include:

- Infection
- Hypoglycaemia
- Stress
- Traumatic injuries
- Surgery
- Burns
- Pregnancy
- Medical emergencies such as MI
- Sudden stop to steroid treatment



(Hahner. and Allolio, 2009).

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Adrenal Insufficiency/Addison's Disease

The body is unable to increase cortisol production to respond to the stress levels and may enter a state of crisis.

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Adrenal Insufficiency/Addison's Disease

- Assess and manage ABC symptomatically
- Maintain SP02 > 94%
- IV access if indicated – by paramedic
- 100mg Hydrocortisone IV or IM – Not in AAP scope of practice
- Manage hypoglycaemia with IV glucose where required – by paramedic
- ECG

Rapid transport to nearest suitable ED facility

(Caroline (2014), JRCALC+, National Institute for Clinical Excellence (2017)

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Adrenal Insufficiency/Addison's Disease

Questions?



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Adrenal Insufficiency/Addison's Disease

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JRCALC +

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